

E-Commerce Sales Analysis and Dashboard Development Using Power BI

Project Objective

The objective of this project is to analyze an E-commerce sales dataset and evaluate business performance using key financial metrics. The goal is to develop an interactive Power BI dashboard that supports data-driven decision-making.

Dataset Description

The dataset contains transactional sales records including Order Date, Region, Product Category, Product Name, Sales, Profit, and Quantity. It enables analysis of revenue trends, regional performance, and product-level profitability.

Data Cleaning

The data was cleaned in Excel by removing duplicates, handling missing values, and correcting data types. Order Date was converted to Date format, and numeric fields such as Sales and Profit were standardized to ensure accurate analysis.

Dashboard Development

The cleaned dataset was imported into Power BI, where DAX measures were created to calculate key KPIs. An interactive dashboard was built using KPI cards, charts, and slicers to visualize sales trends, regional performance, and profitability insights.

Key Business Insights

- Regional sales performance varies significantly, with certain regions contributing a higher share of total revenue, indicating stronger market presence and demand.
- Product categories differ in profitability, with some categories generating higher margins despite moderate sales volumes, highlighting the importance of focusing on both revenue and profit.
- Monthly sales trends reveal seasonal fluctuations, suggesting that customer purchasing behavior is influenced by specific time periods and promotional cycles.

Business Recommendations

- Allocate greater marketing and operational focus toward high-margin product categories and high-performing regions to maximize profitability.
- Utilize seasonal sales trends for strategic planning by aligning inventory management, promotional campaigns, and pricing strategies with peak demand periods.